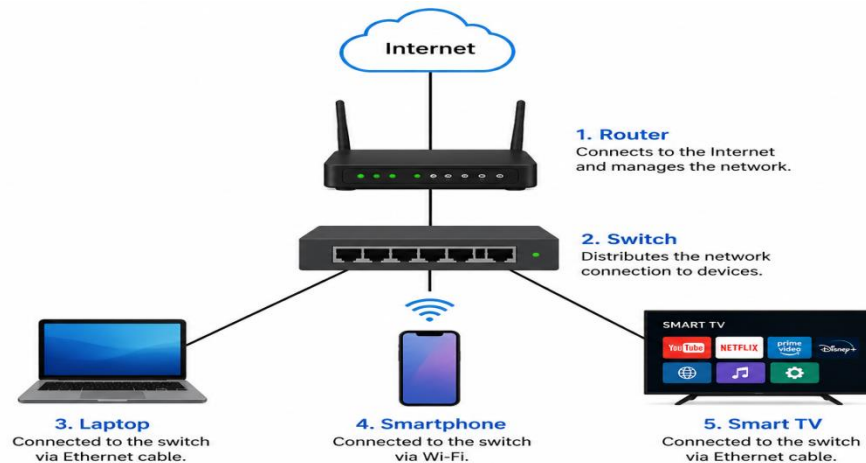


1. Draw and label a simple home network diagram showing a router, switch, and three devices (like a laptop, smartphone, and smart TV), indicating how each device connects to the network.



2. Create a comparison table listing at least four network devices (router, switch, hub, access point) and describe one real-world use case for each, referencing apps or services you use daily (e.g., Wi-Fi for streaming on YouTube, LAN gaming, etc.).

a. Router

- **Primary Function**

- Connect different networks and directs internet traffic.

- **How it works**

- Routes data packets between home network and the internet.

- **Real-World Use Case**

- Sharing one internet connection with all devices at home.

- **Example of Daily App/Service**

- Watching videos on YouTube, browsing websites or attending online meetings over home Wi-Fi.

b. Switch

- **Primary Function**

- Connect devices within the same Local Area Network.

- **How it works**

- Sends data only to the intended device using MAC addresses.

- **Real-World Use Case**

- Connecting multiple desktop PCs in an office or for LAN gaming with low latency.

- **Example of Daily App/Service**

- Multiplayer LAN games, sharing files between computers or accessing a network printer.

c. Hub

- **Primary Function**

- Connect multiple devices in a LAN.

- **How it works**

- Broadcasts incoming data to all connected devices.

- **Real-world Use Case**

- Small test labs or educational demonstrations where simplicity is more important than performance.

- **Example of Daily App/Service**

- Learning networking concepts or connecting a few computers for basic file sharing.

d. Wireless Access Point (WAP)

- **Primary Function**

- Provide Wi-Fi access to a wired network.

- **How it works**

- Converts a wired network connection into a Wi-Fi signal for wireless devices.

- **Real-world Use Case**

- Connecting smartphones, tablets, laptops, and smart TVs to Wi-Fi.

- **Example Daily App/Service**

- Streaming YouTube, using Netflix, browsing social media or making WhatsApp calls over Wi-Fi.

3. Write a short scenario where you explain how data travels from your phone to the internet when you order food on Zomato, mentioning the role of each network device involved (router, switch, and access point).

- When I open Zomato on my phone and place a food order, my phone first connects wirelessly to the access point, which provides the Wi-Fi connection. The access point sends the data to the switch, which forwards it efficiently within the local network to the correct device. The data then reaches the router, which directs it from my home network to the internet. The router sends the request to Zomato's servers, where the order is processed. The confirmation travels back through the internet to the router, then the switch, the access point and finally to my phone, where I can see that my order has been successfully placed.

4. Given this situation: you have a Wi-Fi router, but your gaming PC only has an Ethernet port and is far from the router. Suggest two different network devices you could use to connect your PC to the internet and explain how each would work in this setup.

- Two methods, I would use to explain the setup.

1) Powerline Adapter

- A powerline kit comes with two adapters.
- Plug one adapter into a power outlet near Wi-Fi router and connect it to the router with an Ethernet cable.
- Plug the second adapter into a power outlet near your gaming PC and connect it to the PC with an Ethernet cable.
- The adapters use home's electrical wiring to carry the network signal, giving PC a wired internet connection without running a long Ethernet cable.

2) Wi-Fi Extender with an Ethernet Port

- Place the Wi-Fi extender somewhere between the router and gaming PC where it still receives a strong Wi-Fi signal.
- The extender connects wirelessly to router and rebroadcasts the Wi-Fi signal.
- Connect the gaming PC to the extender using an Ethernet cable.
- In this setup, the extender acts like a wireless bridge, allowing Ethernet-only PC to access the internet through existing Wi-Fi network.